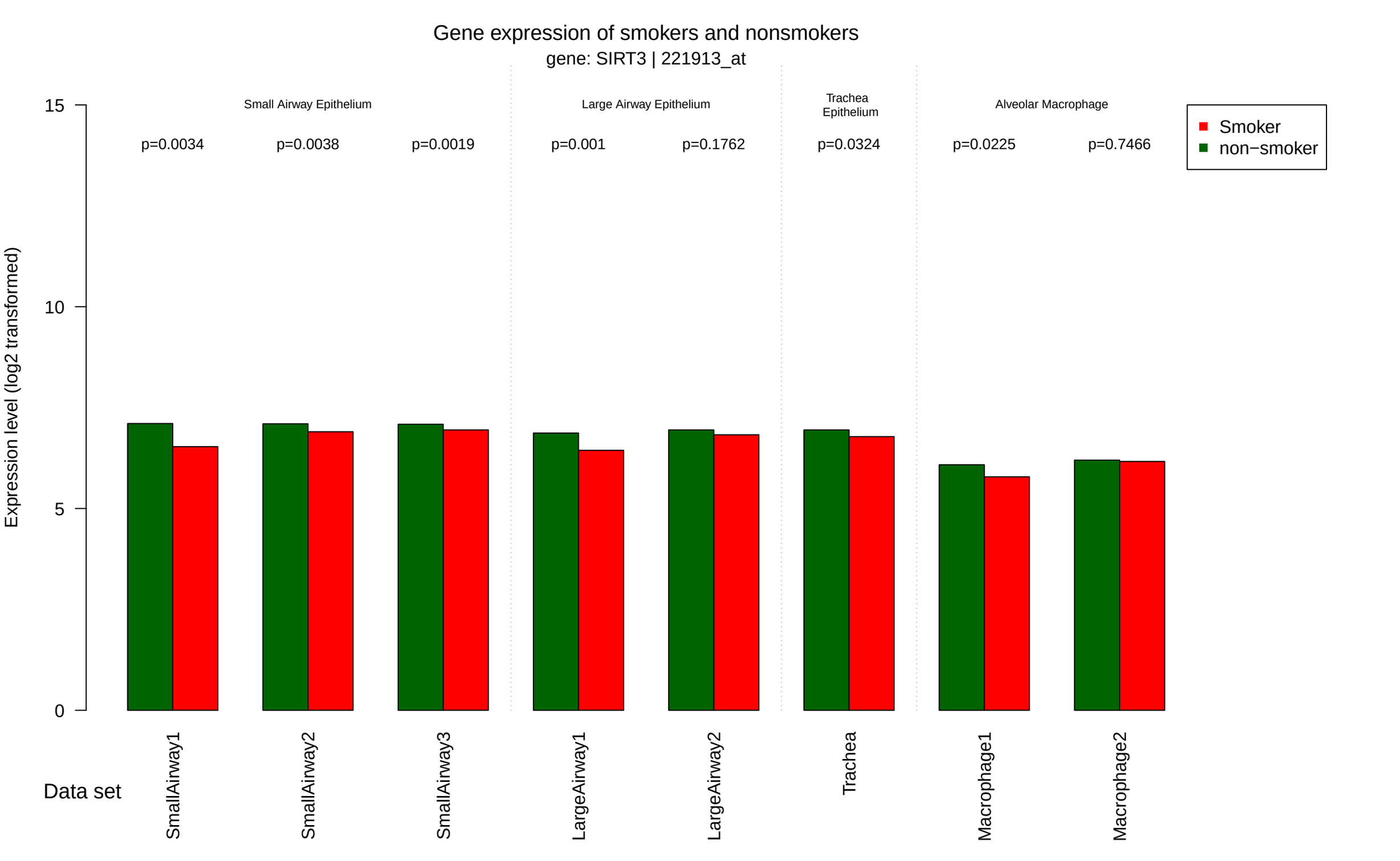
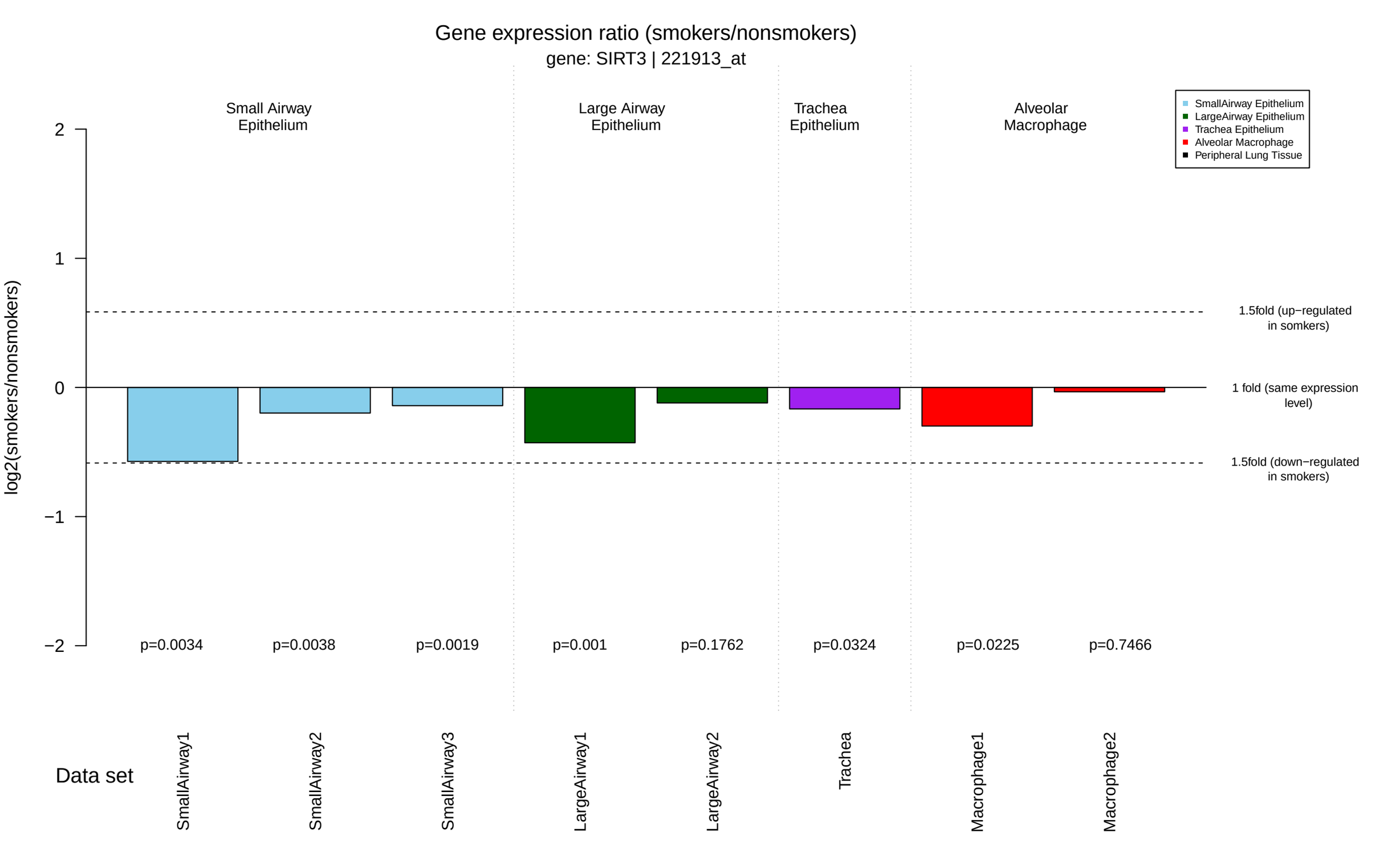
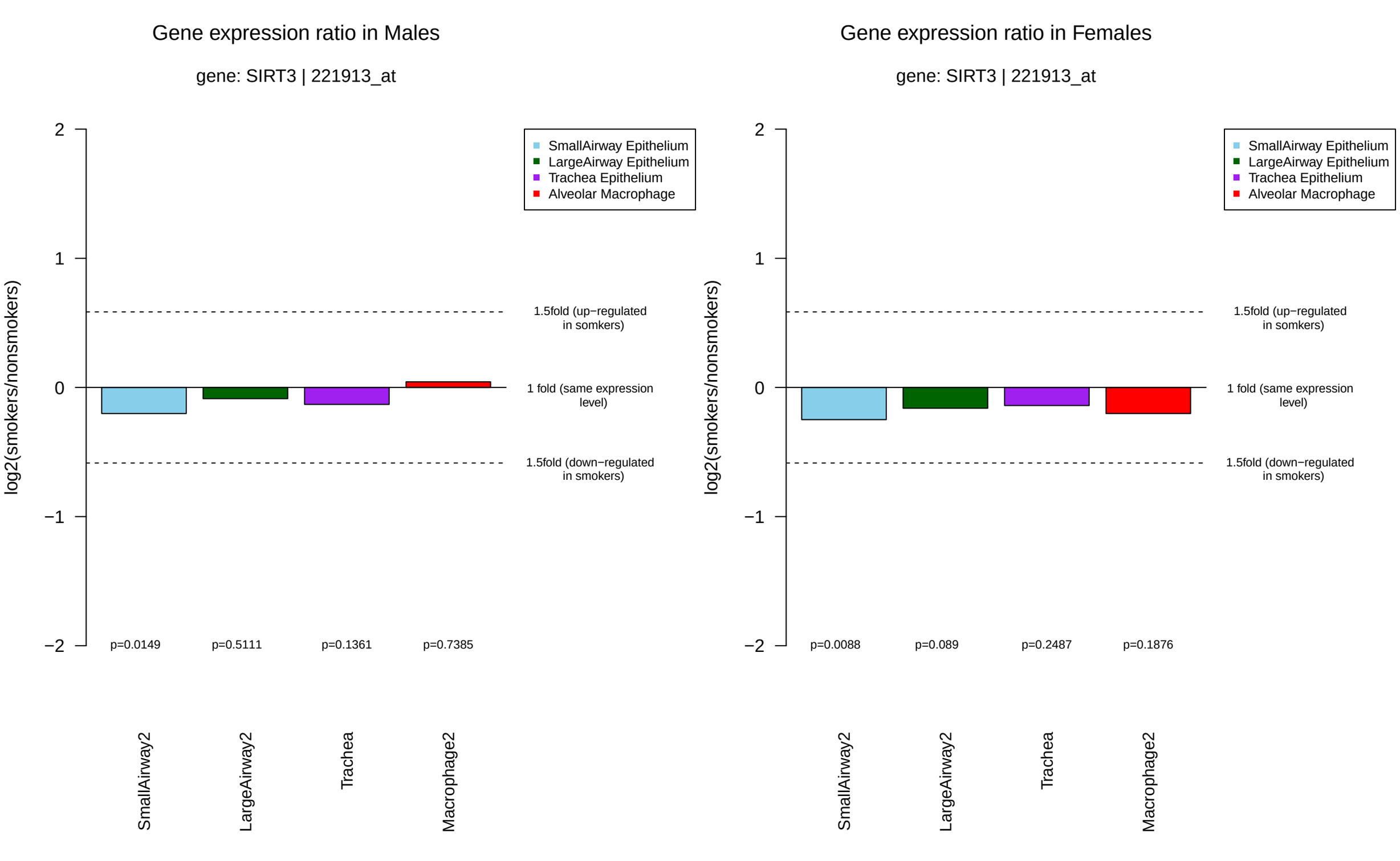
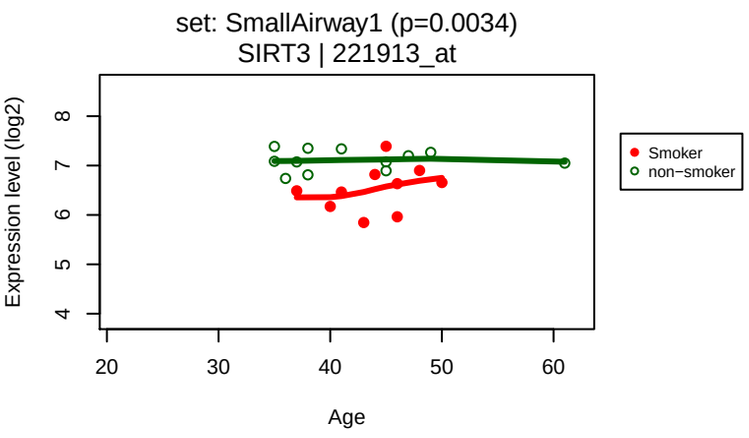
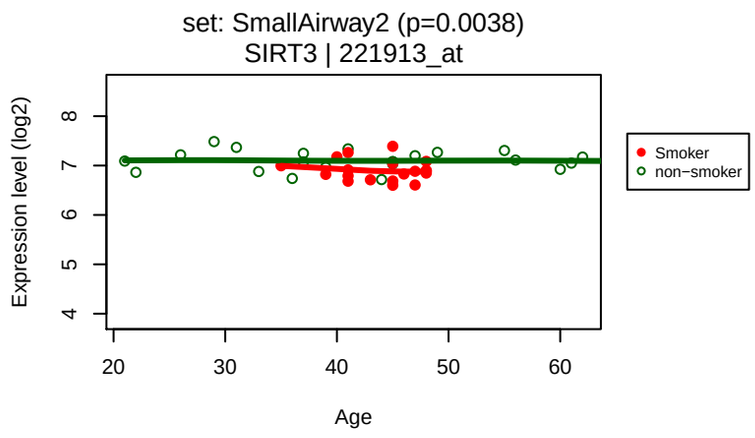
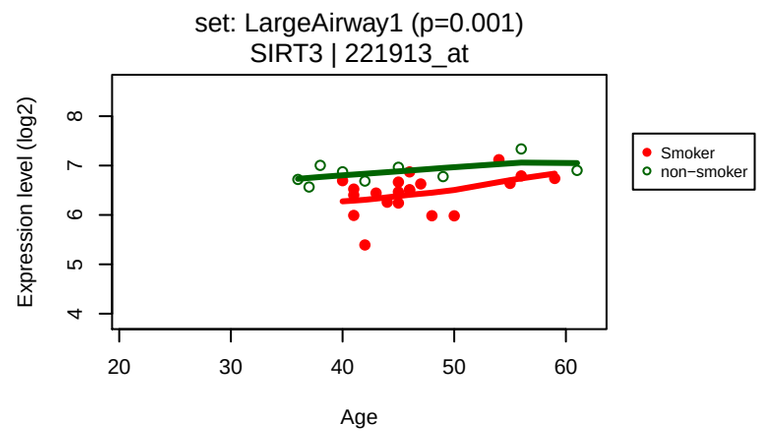

Gene expression of smokers and nonsmokers
gene: SIRT3 | 221913_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0034
p=0.0038
p=0.0019
p=0.001
p=0.1762
p=0.0324
p=0.0225
p=0.7466
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

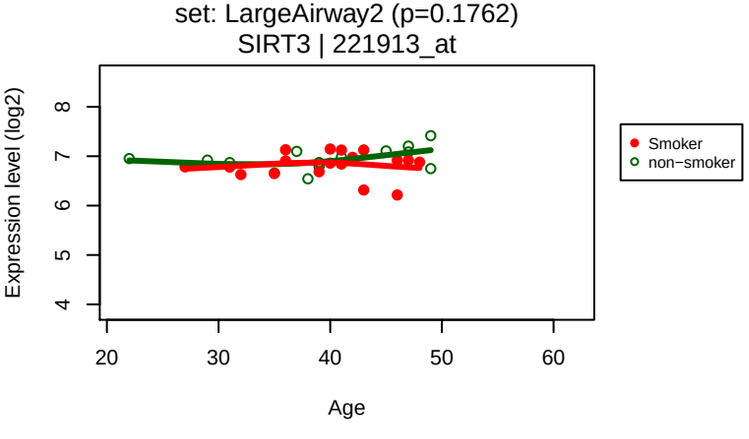

Gene expression ratio (smokers/nonsmokers)
gene: SIRT3 | 221913_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0034
p=0.0038
p=0.0019
p=0.001
p=0.1762
p=0.0324
p=0.0225
p=0.7466
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

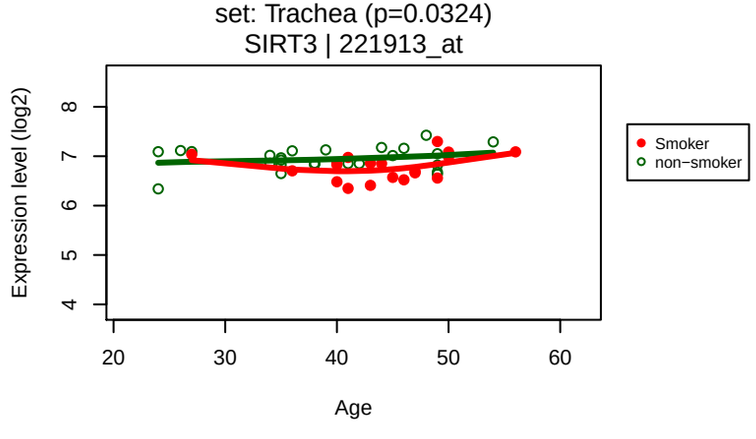

Gene expression ratio in Males
gene: SIRT3 | 221913_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0149
p=0.5111
p=0.1361
p=0.7385
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SIRT3 | 221913_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0088
p=0.089
p=0.2487
p=0.1876
SmallAirway2
LargeAirway2
Trachea
Macrophage2

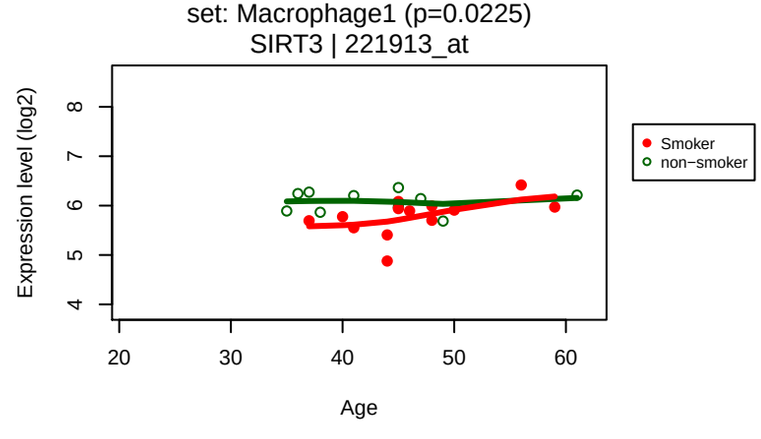

set: SmallAirway1 (p=0.0034)
SIRT3 | 221913_at
Expression level (log2)
Age
Smoker
non−smoker

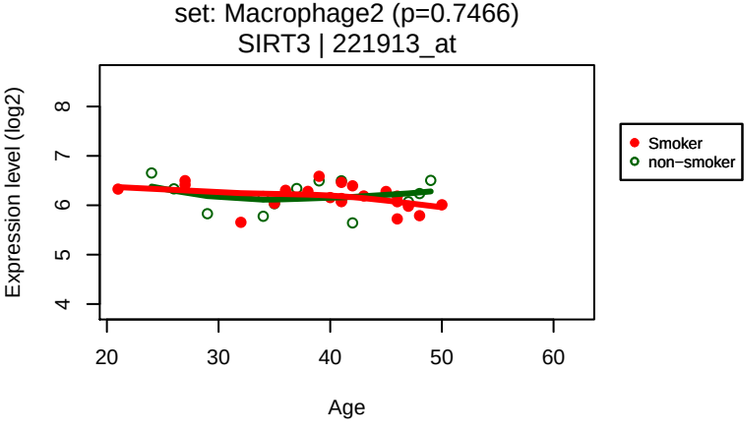

set: SmallAirway2 (p=0.0038)
SIRT3 | 221913_at
Expression level (log2)
Age
Smoker
non−smoker

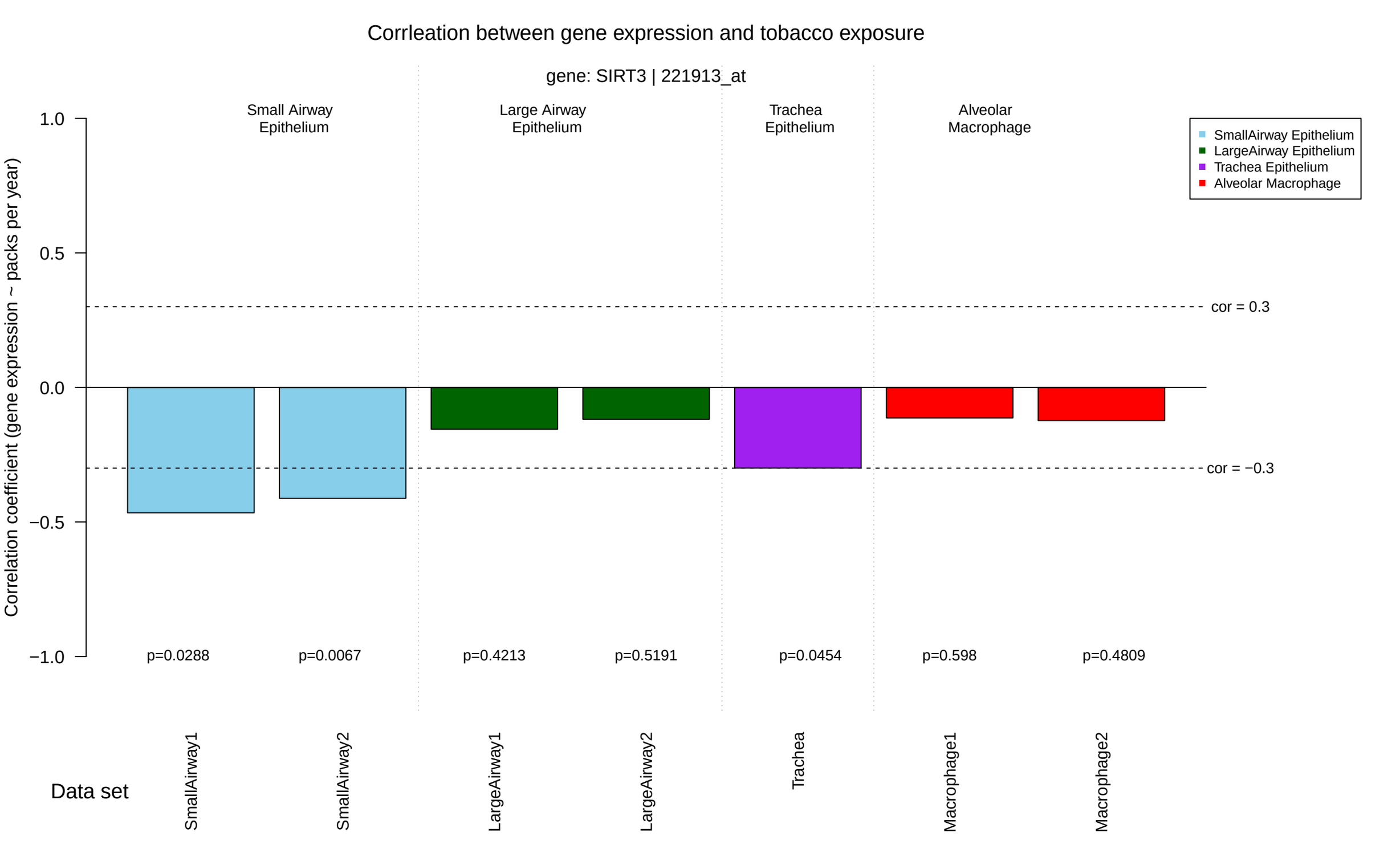

set: LargeAirway1 (p=0.001)
SIRT3 | 221913_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1762)
SIRT3 | 221913_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0324)
SIRT3 | 221913_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0225)
SIRT3 | 221913_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7466)
SIRT3 | 221913_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SIRT3 | 221913_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0288
p=0.0067
p=0.4213
p=0.5191
p=0.0454
p=0.598
p=0.4809
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2